Customer Segmentation using Unsupervised Learning

Applying clustering algorithms to telecom customer data for targeted retention strategies

Why This Matters

High Churn

Telecom companies experience high customer churn which reduces revenue and growth.

Costly Acquisition

Acquiring new customers is 5–7 times more expensive than retaining existing customers.

Personalized Approach

Generic retention strategies fail because customers have different behaviors and needs.

Why Unsupervised Learning?

01

No Labeled Data

No labeled data available for natural customer groups.

02

Discover Patterns

Discovers hidden behavioral patterns.

03

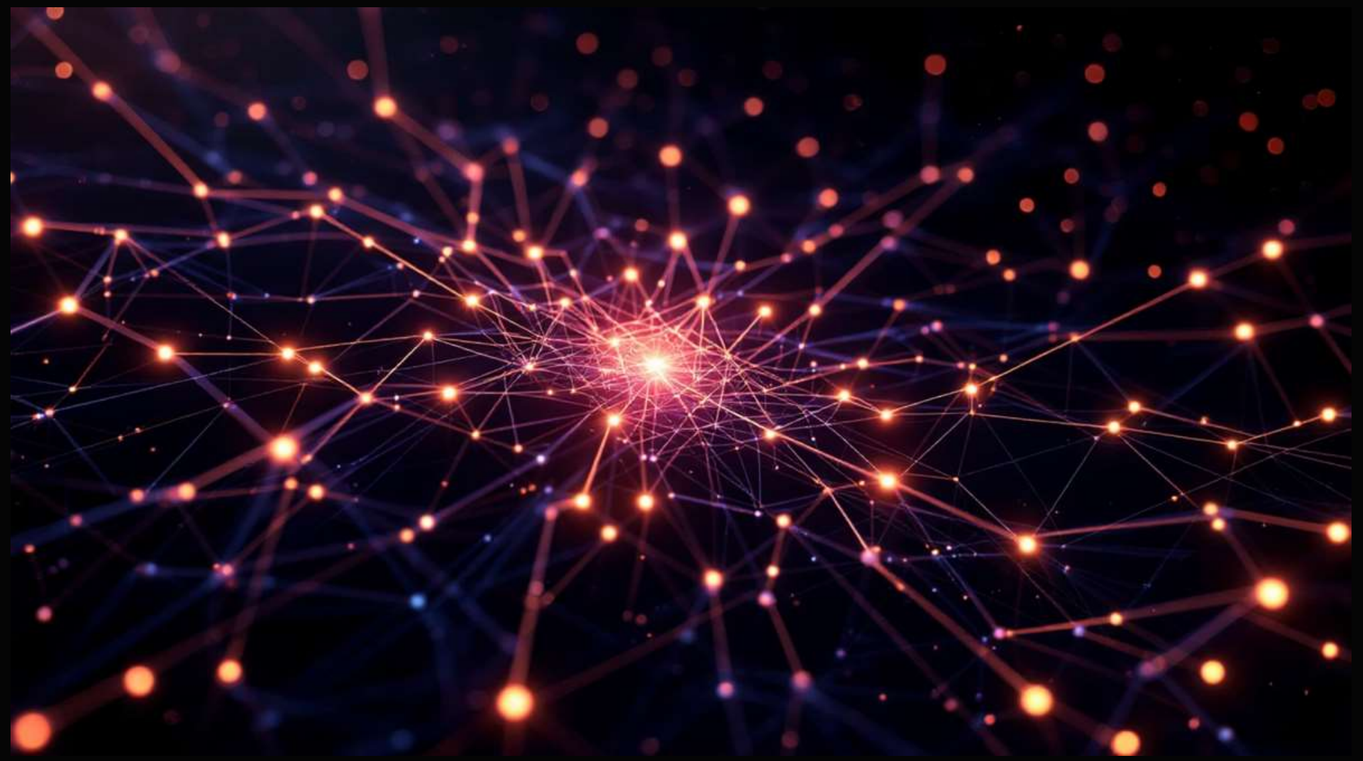
Proactive Strategy

Supports proactive retention strategies.

04

Exploratory Analysis

More exploratory than supervised learning.



Our Approach

Data Collection
Telecom churn dataset with customer behavior metrics

Data Preprocessing
Encoding categorical features and handling missing values

Feature Scaling
Standardization for consistent clustering results

Dimensionality Reduction
PCA for efficient computation and visualization

Clustering
KMeans, DBSCAN, and Hierarchical algorithms

Evaluation
Cluster interpretation and business insights

Algorithm Comparison

KMeans

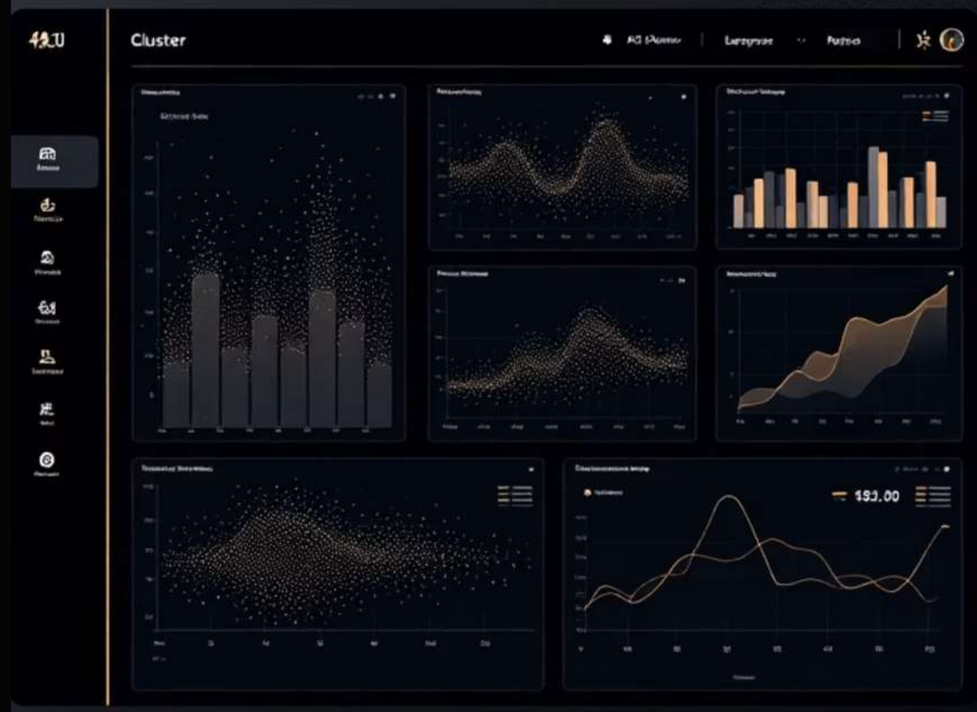
Achieved best performance with strong cluster separation and stability. Silhouette score confirmed as most effective segmentation model.

Hierarchical

Provided useful structural visualization but was computationally heavier than KMeans.

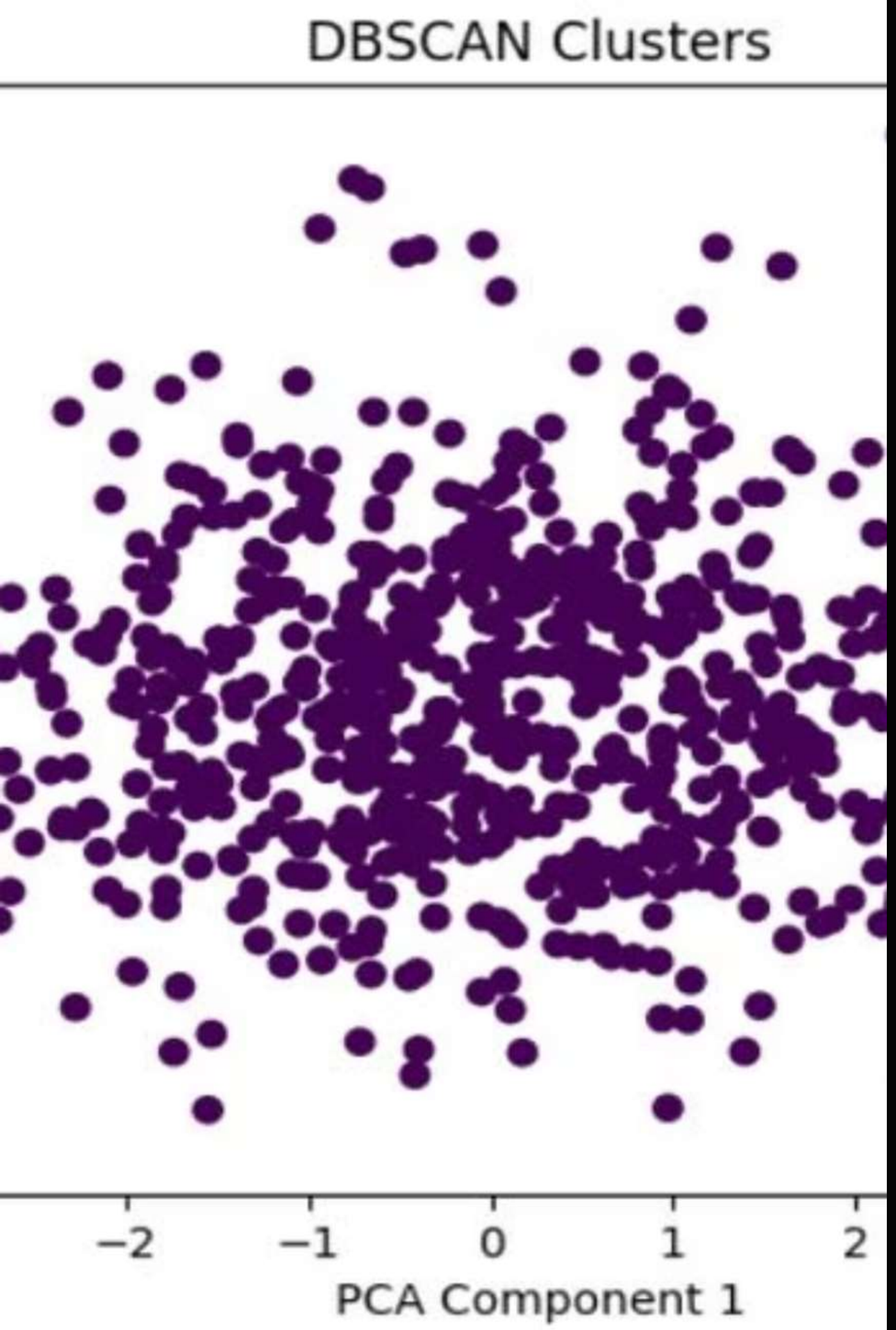
DBSCAN

Identified noise and outliers but produced less consistent cluster boundaries.



KMeans Customer Segments

Three distinct customer segments identified through clustering analysis. Purple cluster centered at approximately (-1, -1), yellow cluster at (0, 2), and teal cluster at (1, 0). Some overlap visible between purple and teal groups.

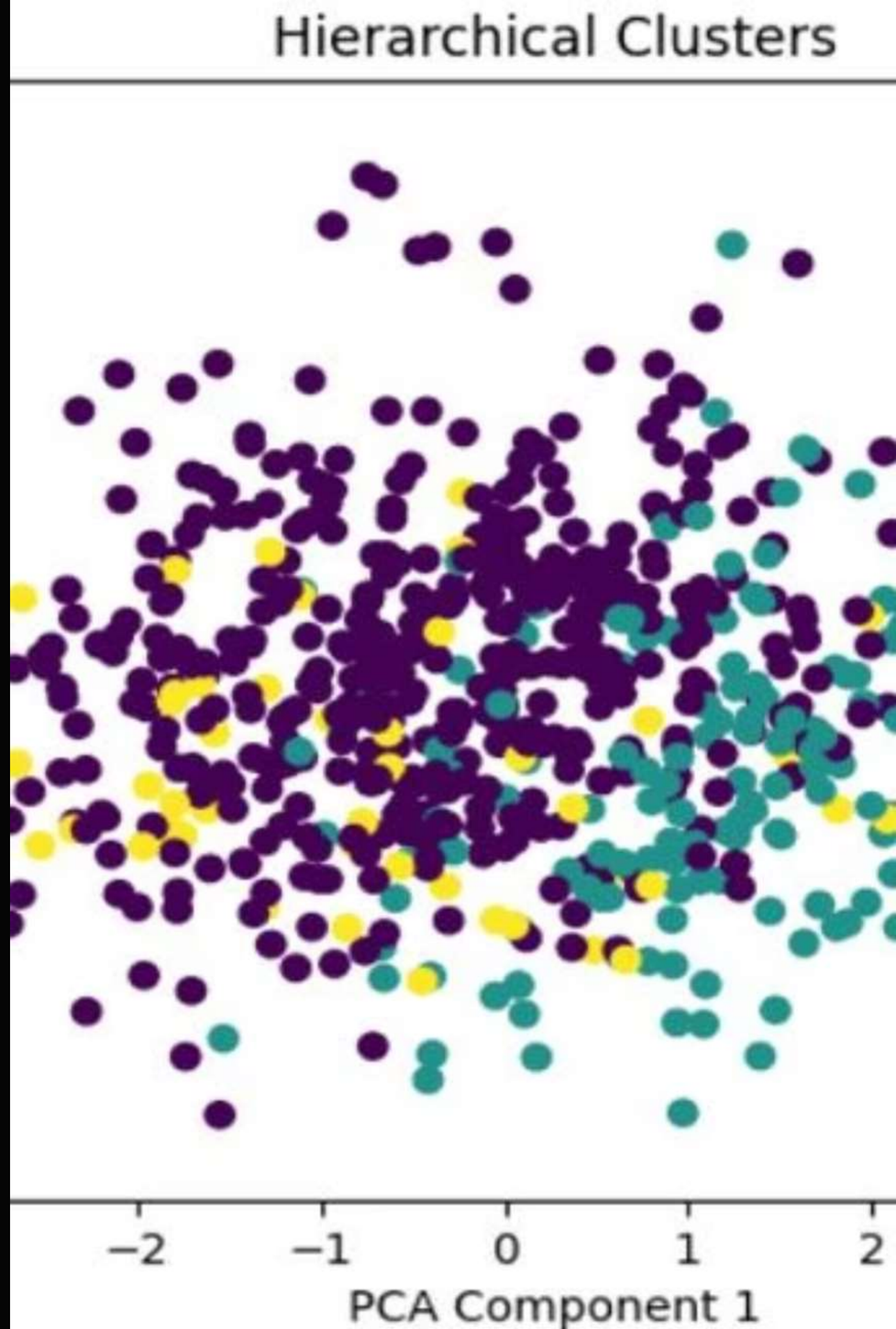


DBSCAN Clusters

Single large cluster with all points colored purple. Points densely packed in center with outliers scattered around periphery. Distribution similar to KMeans plot but without explicit segmentation into multiple clusters.

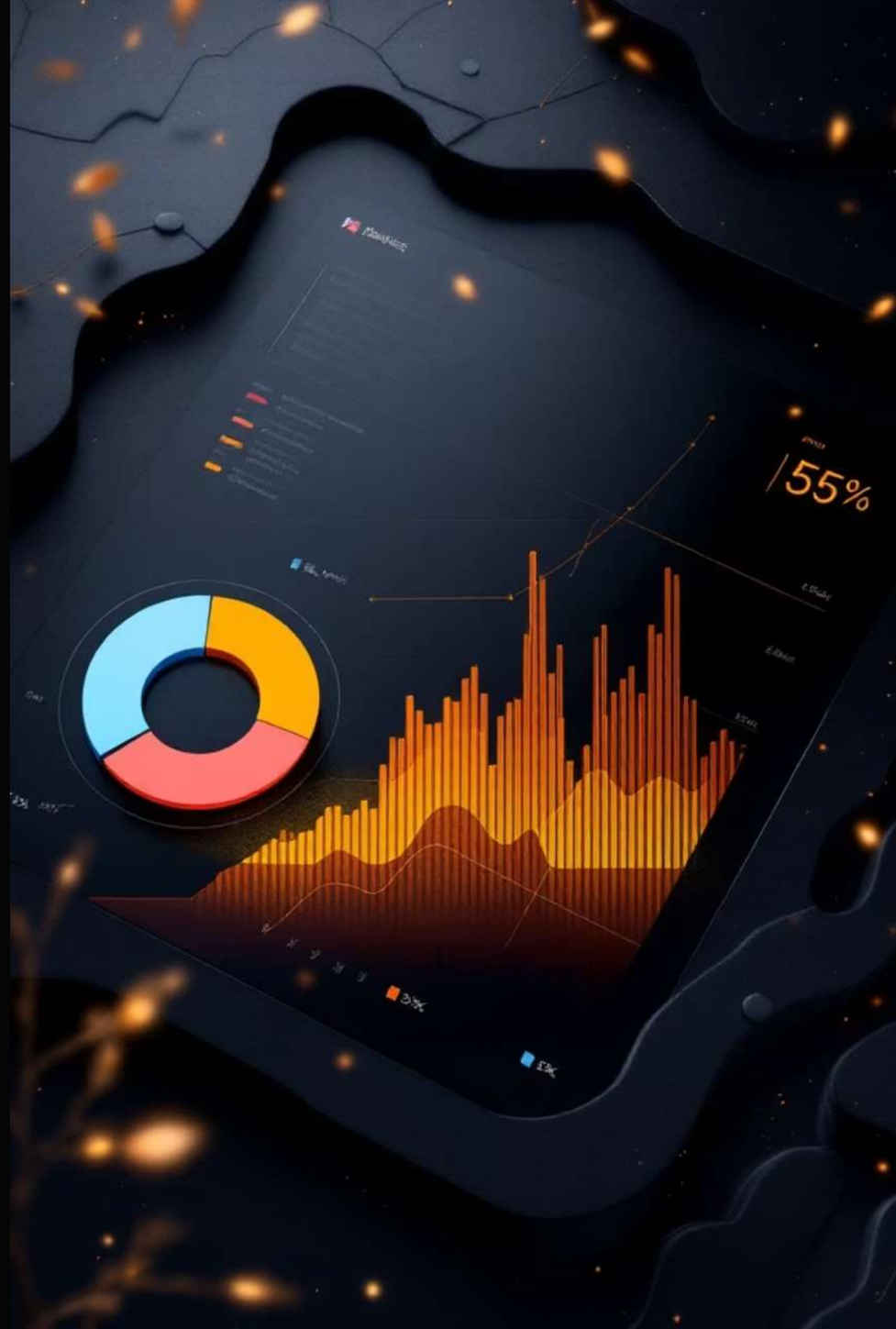
Hierarchical Clusters

Three distinct clusters based on PCA Component 1 and PCA Component 2. Purple cluster centered around (-0.5, 0.5), yellow cluster around (-2.5, -0.5), and teal cluster around (1.5, 0.5).



Cluster Distribution

255	186	226
Cluster 0	Cluster 1	Cluster 2
High-Value Loyal Customers	Price-Sensitive Moderate Users	At-Risk Low Engagement



Business Recommendations

Reward Loyalty

Reward loyal high-value customers to increase lifetime value.

Personalize Offers

Offer personalized discounts to price-sensitive customers.

Proactive Support

Provide proactive support to churn-risk customers.

Improve Service

Improve service quality to reduce complaints.

Targeted Campaigns

Use segmentation for targeted marketing campaigns.

Continuous Learning

Continuously retrain models to improve prediction accuracy.